Name -Vaishnavi U

NicheBot - AI-Powered Community Engagement System

Abstract

NicheBot is an AI-powered conversational assistant designed to enhance discussions in niche online communities. By leveraging Natural Language Processing (NLP) and Machine Learning (ML), NicheBot identifies relevant conversations, generates insightful responses, and fosters engagement among users. The system collects data from RSS feeds, Q&A platforms (Reddit, Discord, Stack Overflow), and niche blog discussions through web scraping. The extracted data is preprocessed and stored in a JSON file for further analysis.Topic modeling is implemented using Latent Dirichlet Allocation (LDA) and Naïve Bayes, allowing the system to categorize discussions accurately. To analyze sentiment, a Sentiment Analysis Model trained on Kaggle's Twitter dataset evaluates the emotional tone behind conversations. These models are stored as .pkl files and vectorized for real-time processing.

A FastAPI-powered backend integrates the AI models with a MongoDB database, enabling users to register, post blogs, and engage in discussions. All comments are monitored by MiniLM-L6-v2, a transformer-based NLP model that generates contextual responses using Together AI's API. This ensures high-quality, meaningful interactions.If a user posts about technology, NicheBot automatically engages, providing relevant responses and encouraging discussions. The chatbot is also integrated with Slack, enabling seamless interaction within workspace communities. The system's accuracy and effectiveness are measured through a user feedback survey, with results stored in a CSV file. The collected metrics power a real-time impact dashboard created using Highcharts.

Tech Stack & Concepts

- **Programming Languages**: Python
- **Frameworks & APIs**: FastAPI, Together AI API
- **Databases**: MongoDB
- **NLP & ML Models**: LDA, Naïve Bayes, MiniLM-L6-v2
- **Sentiment Analysis**: Kaggle-based Twitter dataset
- **Vectorization**: TF-IDF, Count Vectorizer
- **Web Scraping**: BeautifulSoup, Scrapy
- **Data Storage & Visualization**: JSON, CSV, Highcharts
- **Deployment & Integration**: Slack Bot API

2. Features and Implementation

2.1 Discovering Relevant Conversations

- Web scraping implemented to gather data from RSS Feeds, niche community blogs, Q&A platforms (Reddit, Discord, Stack Overflow), and news feeds.
- Extracted data is stored in scraped_data.json for further processing.

2.2 Data Preprocessing and Modeling

- Cleaning Techniques: Tokenization, stopword removal, and lemmatization applied.
- Feature Extraction: TF-IDF and count vectorization used to transform text.
- Topic Modeling: Latent Dirichlet Allocation (LDA) and Naïve Bayes classify discussions into relevant categories.
- Sentiment Analysis: Implemented to analyze emotions behind discussions, using a Twitter dataset from Kaggle.
- Model Training: Finalized models stored as .pkl files along with trained vectorizers.
- Dynamic Testing: Models tested using user-driven inputs via test1.py.

2.3 AI-Powered Community System

- Backend Development: Built using FastAPI and MongoDB for real-time interaction.
- User Features: Users can register, update profiles, post blogs, comment, and engage in discussions.
- Automated Moderation: Comments monitored using NLP models and an AI model (MiniLM-L6-v2 transformer).
- Response Generation: Uses together_ai's API to generate meaningful responses based on contextual embeddings.

2.4 Intelligent Chatbot Integration

- Technology-Related Query Handling: NicheBot automatically responds to discussions related to technology.
- Slack Integration: Created a dedicated Slack channel (NicheBot) to provide responses in real-time with high accuracy.
- User Engagement: Enhances collaboration by facilitating AI-assisted discussions.

2.5 Performance Measurement and Analytics

- User Feedback: Surveys prompt users to rate the quality of responses.
- Data Storage: Survey results are stored in a CSV file.
- Impact Visualization: A dashboard powered by Highcharts displays response impact, engagement trends, and user satisfaction metrics.

3. Unique Selling Points (USP)

3.1 AI-Enhanced Community Engagement

- Unlike traditional discussion platforms, NicheBot provides real-time AI-powered insights and responses to enhance user interactions.

3.2 Automated Moderation & Response Generation

- NLP-based automated content moderation ensures discussions remain relevant and valuable.
- MiniLM-L6-v2 delivers contextually aware, high-quality responses.

3.3 Cross-Platform Integration

- NicheBot is not limited to a single platform; it can be integrated into Slack, Discord, and other community forums.

3.4 Data-Driven Insights

- The system continuously learns from user interactions, improving response accuracy over time.
- Highcharts-powered analytics dashboard provides actionable insights into user engagement.

4. Stakeholders

- Community Users: Bloggers, developers, researchers, and technology enthusiasts looking for valuable discussions.
- Companies & Startups: Organizations aiming to enhance user engagement within their platforms.
- Forum Administrators: Moderators seeking AI-powered assistance in managing discussions.
- Data Scientists & AI Engineers: Professionals interested in AI-driven conversation analysis.

5. Business Model

5.1 Revenue Streams

- Subscription-Based Model: Premium access for businesses integrating NicheBot into their platforms.
- API Access Fees: Companies pay to access AI-powered response APIs.
- Enterprise Solutions: Custom AI-powered engagement solutions for large-scale forums.
- Advertising & Partnerships: Monetization via partnerships with tech blogs and discussion platforms.

5.2 Market Strategy

- Target Audience: Developers, AI researchers, online communities, and corporate engagement teams.
- Marketing Channels: Tech conferences, AI community meetups, LinkedIn campaigns, and developer forums.
- Scalability: Expand to other domains like healthcare, finance, and education by fine-tuning AI models.

6. Conclusion

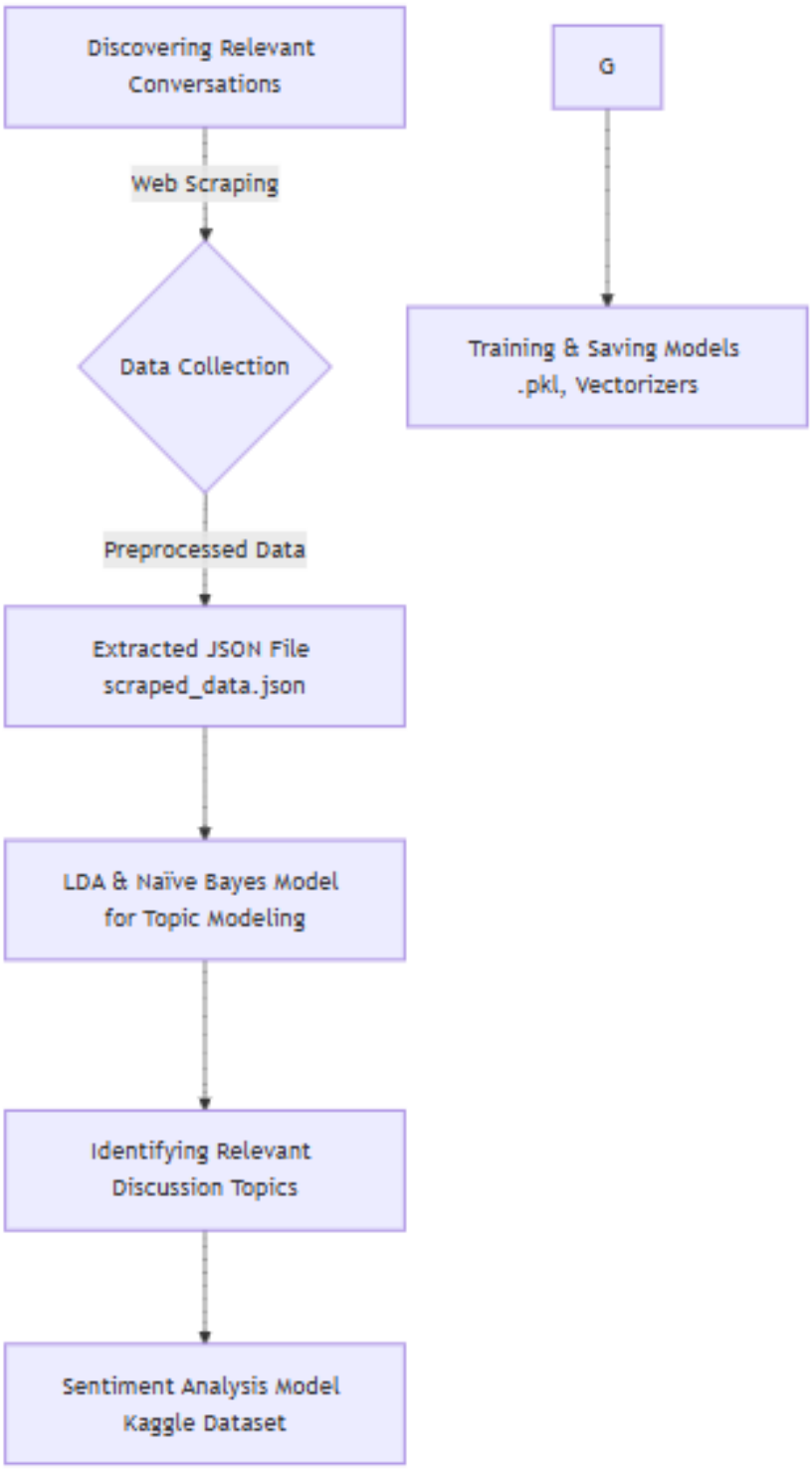
NicheBot represents a significant advancement in AI-assisted community engagement, seamlessly integrating **natural language processing, sentiment analysis, and topic modeling** to enhance user interactions. By leveraging **machine learning models** such as **LDA, Naïve Bayes, and MiniLM-L6-v2**, the system can accurately identify discussion topics, analyze user sentiment, and generate insightful responses.

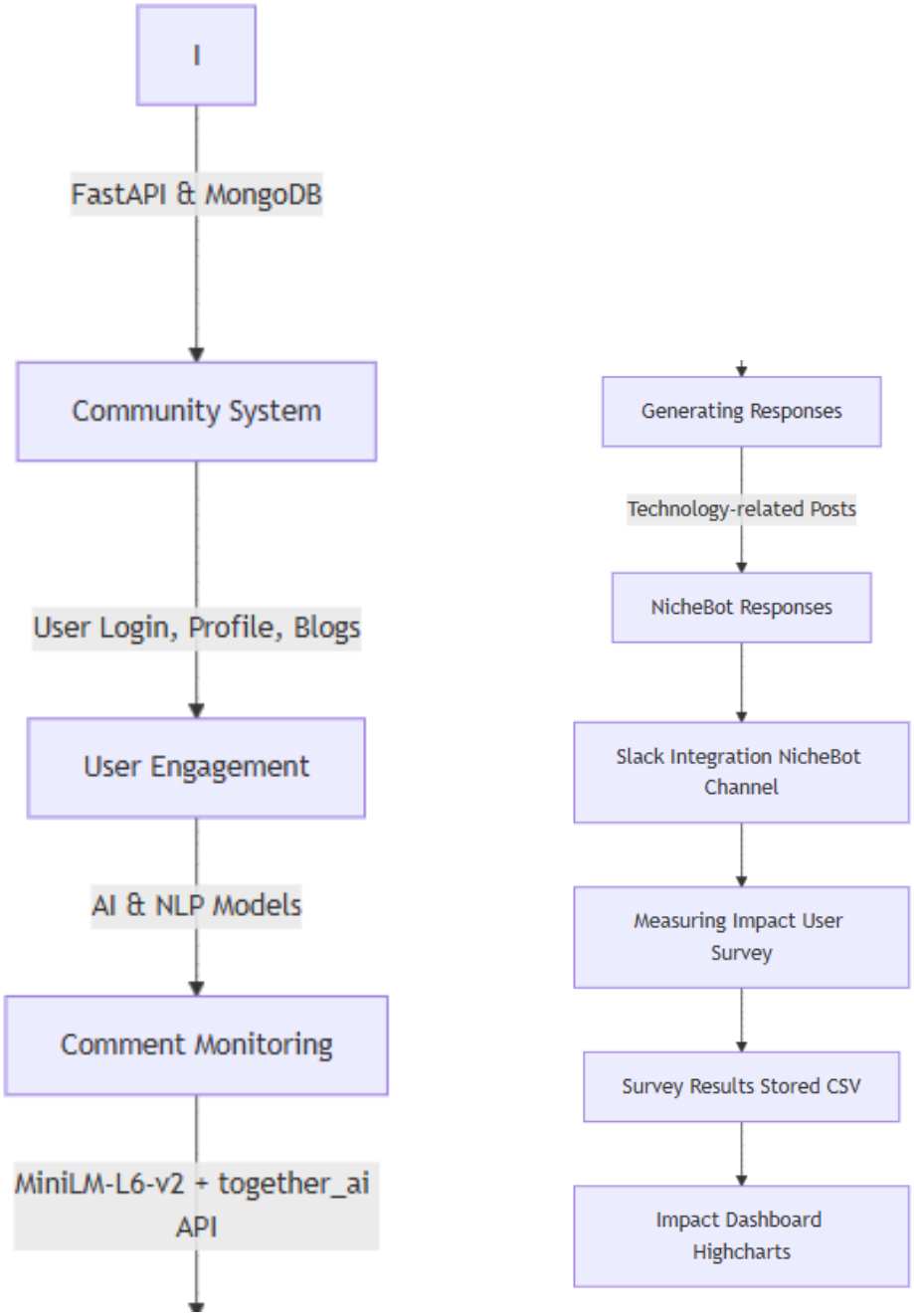
One of the core strengths of NicheBot lies in its ability to **bridge the gap between AI-driven insights and meaningful conversations**. Traditional online forums and communities often struggle with moderation, irrelevant discussions, and lack of engagement. NicheBot effectively addresses these challenges by providing **automated content moderation, intelligent response generation, and real-time topic classification**. This ensures that users receive valuable, context-aware replies, fostering an **interactive and knowledge-driven environment**.

The integration with **Slack** expands its capabilities beyond standalone forums, allowing businesses, developers, and technology enthusiasts to receive instant AI-generated insights within their workspace discussions. Moreover, its backend, built using **FastAPI and MongoDB**, ensures a **scalable, high-performance architecture** capable of handling large volumes of community interactions.

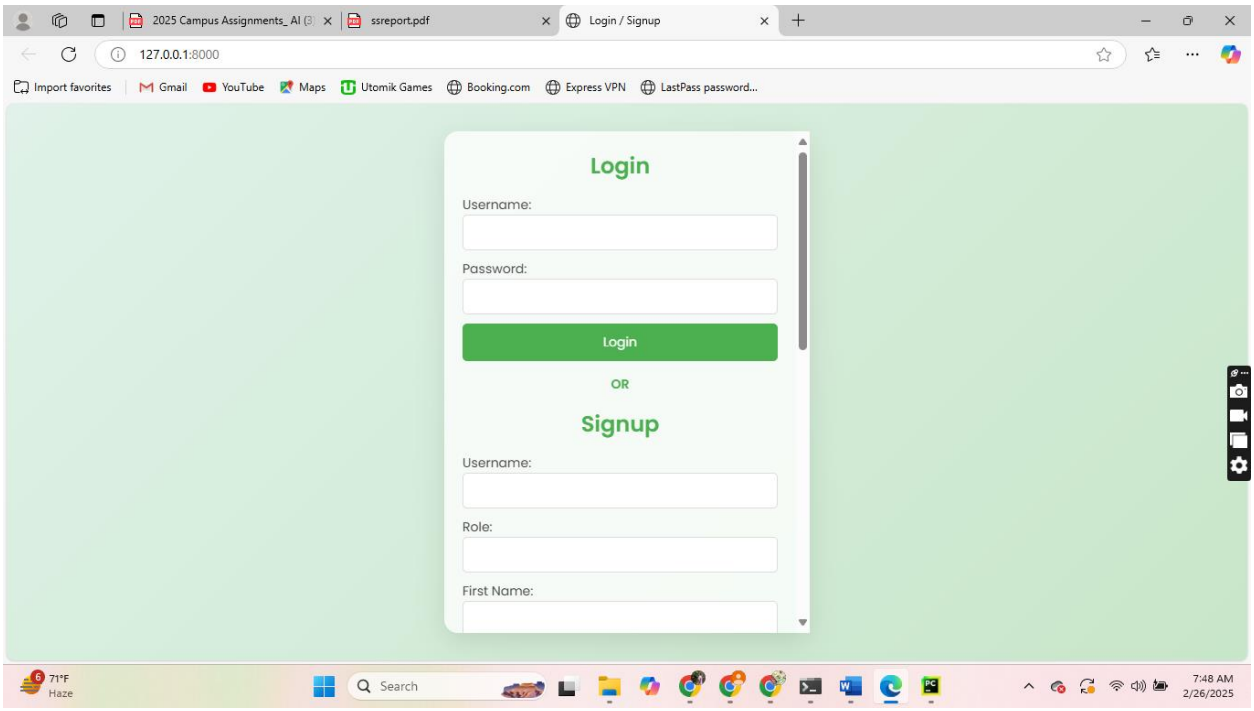
The **impact measurement system**, which collects user feedback and visualizes response effectiveness through **Highcharts**, is another key differentiator. By continuously analyzing survey results, NicheBot improves its conversational abilities over time, **adapting to the evolving needs of users**. This self-improving mechanism makes it a **dynamic and future-proof AI assistant**.

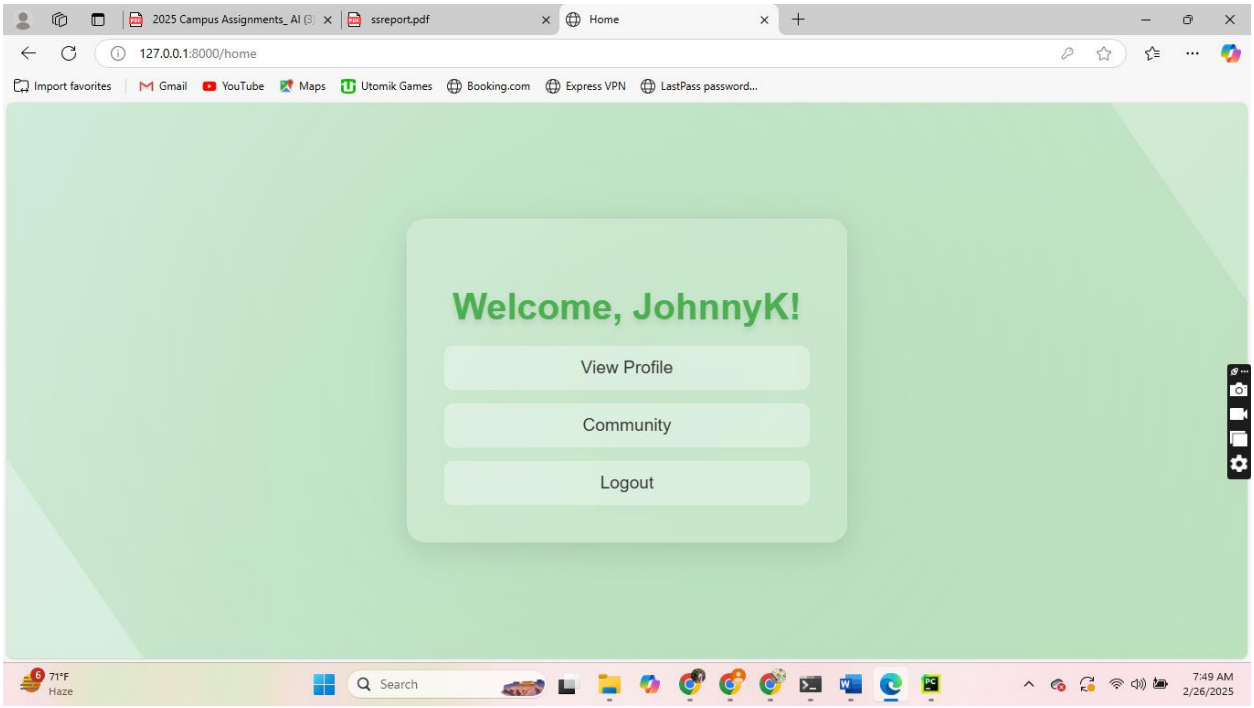
Flow Diagrams:




I
FastAPI & MongoDB
Community System
User Login, Profile, Blogs
User Engagement
AI & NLP Models
Comment Monitoring
MiniLM-L6-v2 + together_ai API
Generating Responses
Technology-related Posts
NicheBot Responses
Slack Integration NicheBot Channel
Measuring Impact User Survey
Survey Results Stored CSV
Impact Dashboard Highcharts

Screenshots of Use Cases:





DBMS Programming: A Guide for Developers

Introduction to DBMS Programming Database Management Systems (DBMS) are at the core of modern applications, handling data storage, retrieval, and manipulation efficiently. Whether you're developing a web app, a mobile application, or an enterprise system, understanding DBMS programming is crucial for optimal performance and scalability. Types of Database Management Systems DBMS can be categorized into different types based on their structure and functionality: Relational DBMS (RDBMS) – Uses structured tables with relationships (e.g., MySQL, PostgreSQL, SQL Server). NoSQL DBMS – Designed for unstructured or semi-structured data (e.g., MongoDB, Cassandra, Redis). NewSQL DBMS – Combines the best of both relational and NoSQL databases (e.g., Google Spanner, CockroachDB).

By: Zade | 2025-02-26 01:00:16

Replied by Google Bot

Comment:

Write a comment...

Add Comment

Comments:

Google Bot: DBMS Programming is a crucial skill for modern application development. It involves understanding the different types of DBMS, such as Relational, NoSQL, and NewSQL, and their respective structures and functionalities. Knowing how to program and manage databases efficiently can greatly improve performance and scalability in web, mobile, and enterprise applications. (2025-02-26 06:33:28)

Evaluate Response

2025 Campus Assignments_ AI (3) | ssreport.pdf | Survey

127.0.0.1:8000/survey/67be1a48f597b37d41939832/67be686032ac624ecee8d234

Import favorites | Gmail | YouTube | Maps | Utomik Games | Booking.com | Express VPN | LastPass password...

Survey: Evaluate Google Bot's Response

Was the response relevant to the question?

1 - Strongly Disagree

Did the response provide accurate information?

1 - Strongly Disagree

Was the response easy to understand?

1 - Strongly Disagree

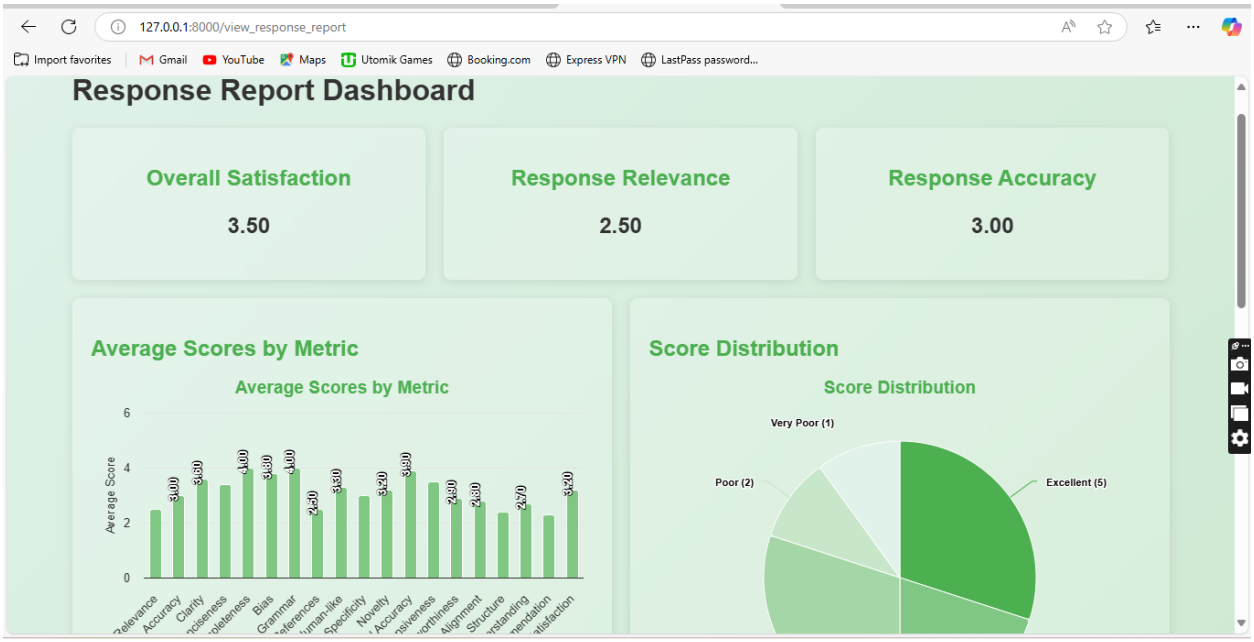
Did the response include unnecessary information?

1 - Strongly Disagree

Was the response concise?

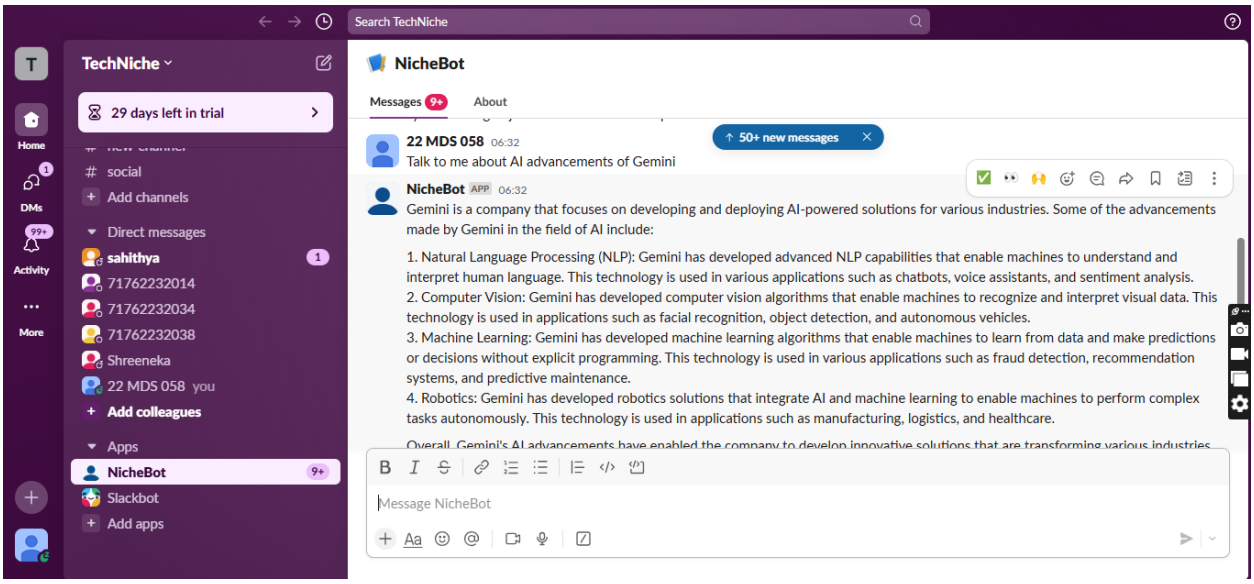
1 - Strongly Disagree

73°F Haze | Search | 7:57 AM 2/26/2025



Recommendations

- Improve Relevance (Current Score: 2.50)
- Improve References (Current Score: 2.50)
- Improve Trustworthiness (Current Score: 2.90)
- Improve Expectation Alignment (Current Score: 2.80)
- Improve Structure (Current Score: 2.40)
- Improve Understanding (Current Score: 2.70)
- Improve Recommendation (Current Score: 2.30)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29 days left in trial
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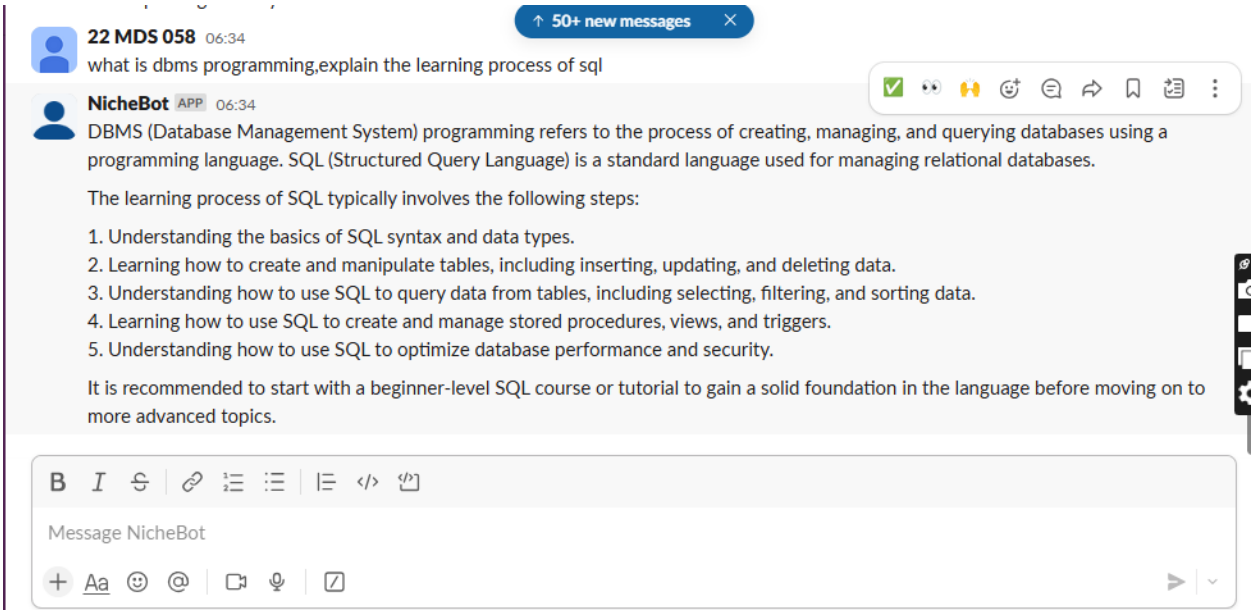
71762232038
Shreeneka
22 MDS 058 you
Add colleagues
Apps
NicheBot
Slackbot
Add apps
NicheBot
Messages
About
50+ new messages
22 MDS 058 06:32
Talk to me about AI advancements of Gemini
NicheBot APP 06:32
Gemini is a company that focuses on developing and deploying AI-powered solutions for various industries. Some of the advancements made by Gemini in the field of AI include:
1. Natural Language Processing (NLP): Gemini has developed advanced NLP capabilities that enable machines to understand and interpret human language. This technology is used in various applications such as chatbots, voice assistants, and sentiment analysis.
2. Computer Vision: Gemini has developed computer vision algorithms that enable machines to recognize and interpret visual data. This technology is used in applications such as facial recognition, object detection, and autonomous vehicles.
3. Machine Learning: Gemini has developed machine learning algorithms that enable machines to learn from data and make predictions or decisions without explicit programming. This technology is used in various applications such as fraud detection, recommendation systems, and predictive maintenance.
4. Robotics: Gemini has developed robotics solutions that integrate AI and machine learning to enable machines to perform complex tasks autonomously. This technology is used in applications such as manufacturing, logistics, and healthcare.
Message NicheBot


50+ new messages
22 MDS 058 06:34
what is dbms programming,explain the learning process of sql
NicheBot APP 06:34
DBMS (Database Management System) programming refers to the process of creating, managing, and querying databases using a programming language. SQL (Structured Query Language) is a standard language used for managing relational databases.
The learning process of SQL typically involves the following steps:
1. Understanding the basics of SQL syntax and data types.
2. Learning how to create and manipulate tables, including inserting, updating, and deleting data.
3. Understanding how to use SQL to query data from tables, including selecting, filtering, and sorting data.
4. Learning how to use SQL to create and manage stored procedures, views, and triggers.
5. Understanding how to use SQL to optimize database performance and security.
It is recommended to start with a beginner-level SQL course or tutorial to gain a solid foundation in the language before moving on to more advanced topics.
Message NicheBot

5:57 100%

NicheBot

Messages About

sahithya 22:21
Helloo

Today

sahithya 05:57
What is ai

NicheBot APP 05:57
AI stands for Artificial Intelligence, which refers to the development of computer systems that can perform tasks that typically require human intelligence, such as learning, problem-solving, and decision-making. AI systems use algorithms and machine learning techniques to process and analyze data, and can be programmed to perform a wide range of tasks, from simple automation to complex decision-making. AI is a rapidly growing field with applications in many areas, including healthcare, finance, transportation, and education.

Message NicheBot